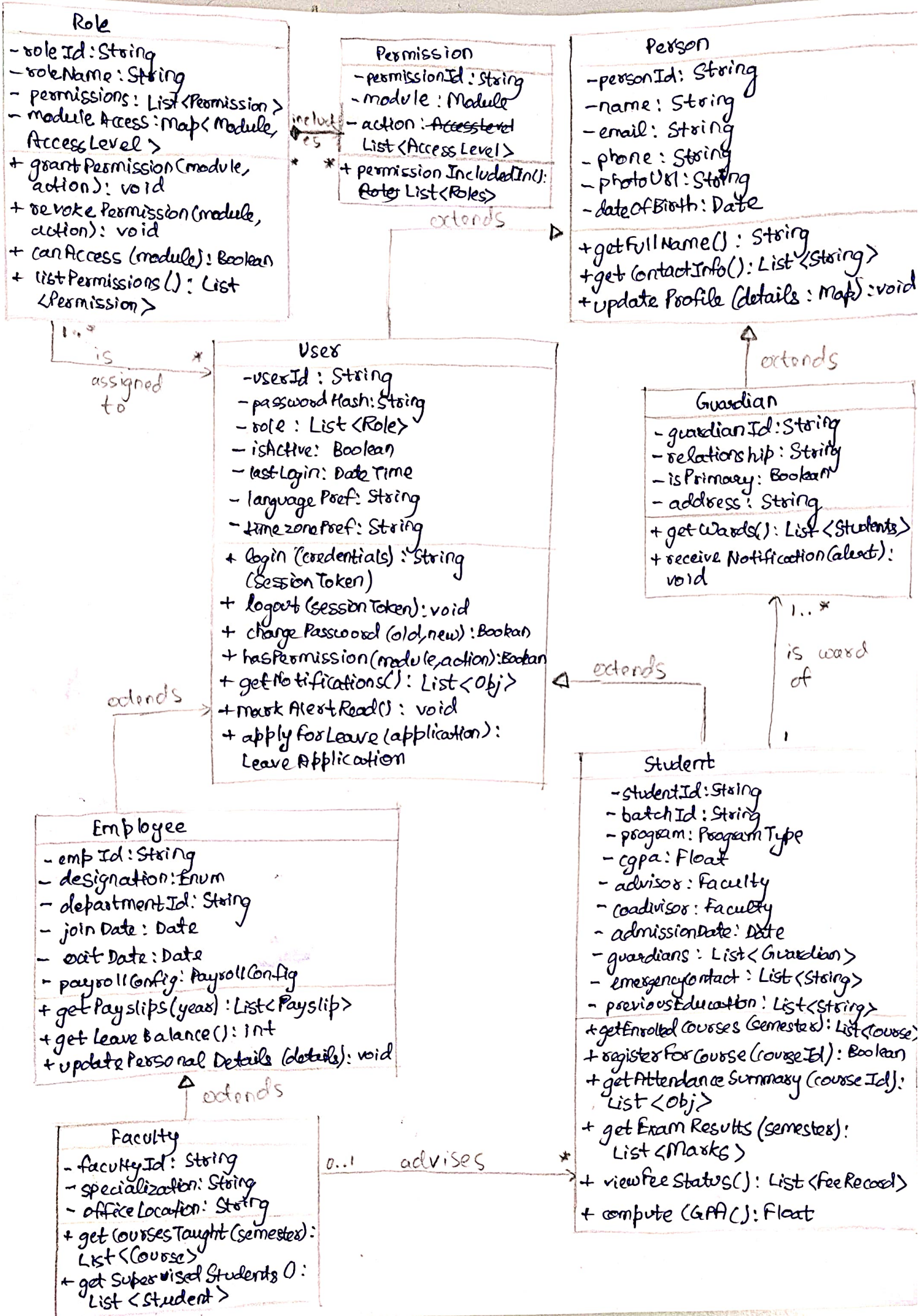
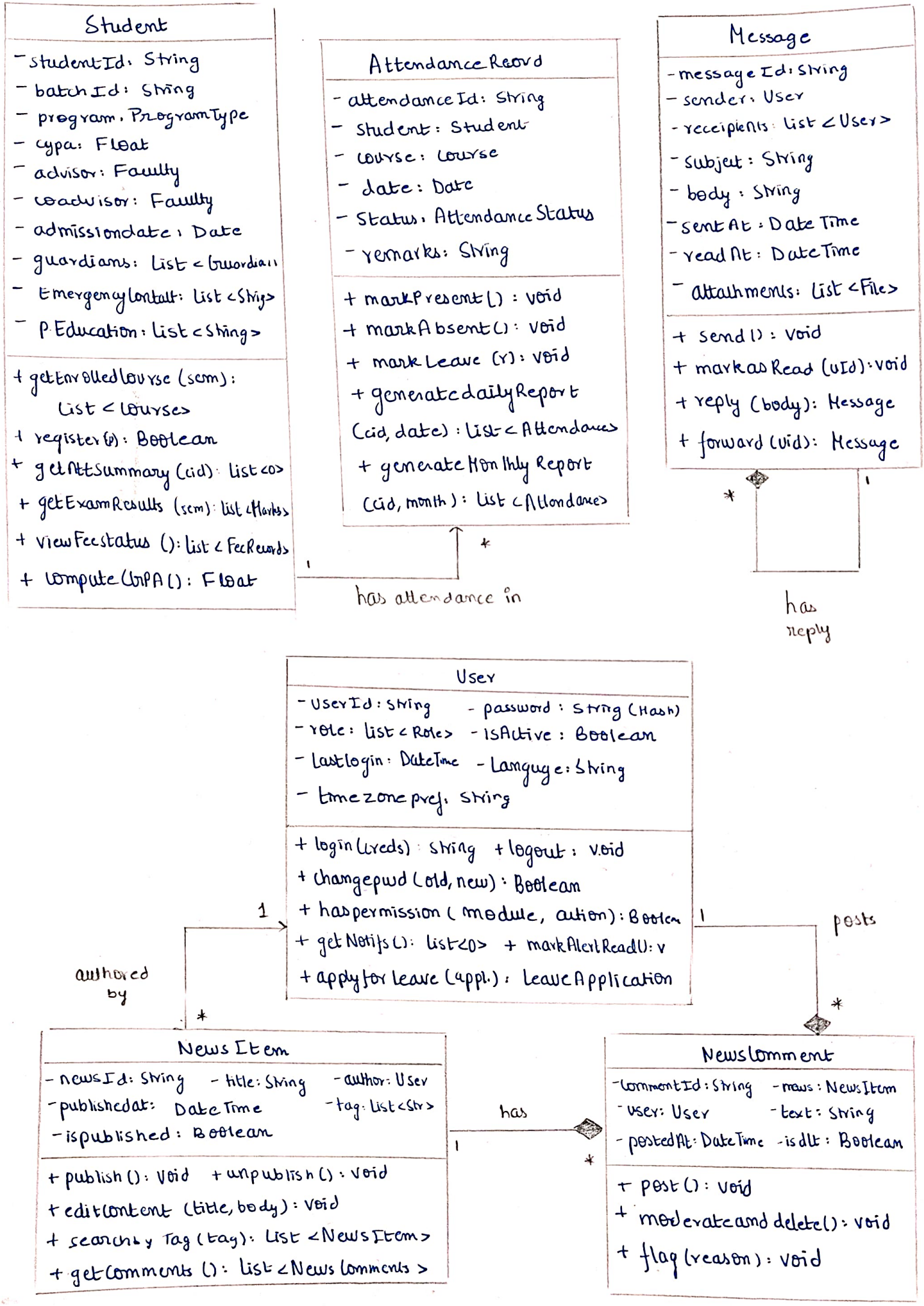

Role
- roleId : String
- roleName : String
- permissions : List<Permission>
- moduleAccess : Map<Module, AccessLevel>
+ grantPermission(module, action) : void
+ revokePermission(module, action) : void
+ canAccess(module) : Boolean
+ listPermissions() : List<Permission>
Permission
- permissionId : String
- module : Module
- action : ~~AccessLevel~~ List<AccessLevel>
+ permissionIncludedIn() : ~~Role~~ List<Roles>
includes
*
*
Person
- personId : String
- name : String
- email : String
- phone : String
- photoUrl : String
- dateOfBirth : Date
+ getFullName() : String
+ getContactInfo() : List<String>
+ updateProfile(details : Map) : void
extends
1..*
is assigned to
*
User
- userId : String
- passwordHash : String
- role : List<Role>
- isActive : Boolean
- lastLogin : DateTime
- languagePref : String
- timezonePref : String
+ login(credentials) : String (SessionToken)
+ logout(sessionToken) : void
+ changePassword(old, new) : Boolean
+ hasPermission(module, action) : Boolean
+ getNotifications() : List<Obj>
+ markAlertRead() : void
+ applyForLeave(application) : LeaveApplication
extends
Guardian
- guardianId : String
- relationship : String
- isPrimary : Boolean
- address : String
+ getWards() : List<Students>
+ receiveNotification(alert) : void
extends
1..*
is ward of
1
extends
extends
Employee
- empId : String
- designation : Enum
- departmentId : String
- joinDate : Date
- exitDate : Date
- payrollConfig : PayrollConfig
+ getPayslips(year) : List<Payslip>
+ getLeaveBalance() : int
+ updatePersonalDetails(details) : void
extends
Faculty
- facultyId : String
- specialization : String
- officeLocation : String
+ getCoursesTaught(semester) : List<Course>
+ getSupervisedStudents() : List<Student>
0..1
advises
*
Student
- studentId : String
- batchId : String
- program : ProgramType
- cgpa : Float
- advisor : Faculty
- coadvisor : Faculty
- admissionDate : Date
- guardians : List<Guardian>
- emergencyContact : List<String>
- previousEducation : List<String>
+ getEnrolledCourses(semester) : List<Course>
+ registerForCourse(courseId) : Boolean
+ getAttendanceSummary(courseId) : List<Obj>
+ getExamResults(semester) : List<Marks>
+ viewFeeStatus() : List<FeeRecord>
+ compute(GPA() : Float

User
- userId : String
- passwordHash : String
- role : List <Role>
- isActive : Boolean
- lastLogin : DateTime
- languagePref : String
- timezonePref : String
+ login (credentials) : String (SessionToken)
+ logout (sessionToken) : void
+ changePassword (old, new) : Boolean
+ hasPermission (module, action) : Boolean
+ getNotifications() : List <obj>
+ markAlertRead() : void
+ applyForLeave (application) : LeaveApplication
has obligations of
1
*
FeeRecord
- feeId : String
- student : Student
- category : FeeCategory
- amount : Float
- dueDate : Date
- paidDate : Date
- status : PaymentStatus
- receiptNumber : String
+ processPayment (amount) : String
+ issueRefund (reason) : void
+ flagAsDefaulter() : void
+ generateReceipt() : String
Recruitment
- interviewId : String
- candidate : Person
- mode : InterviewMode
- panelists : List <User>
- scheduledDate : DateTime
- outcome : InterviewOutcome
+ scheduleInterview (date, mode) : void
+ assignPanel (facultyIds) : void
+ recordOutcome (result) : void
+ sendInvitation() : void
* 1
is panelist for
1
applies for
*
LeaveApplication
- leaveId : String
- applicant : User
- leaveType : Enum
- fromDate : Date
- toDate : Date
+ submit() : void
+ cancel() : void
+ getDuration() : Int
+ deductFromBalance() : Int
1
Interview for
1
tracks expenditure as
*
Transaction
- txnId : String
- type : TxnType
- amount : Float
- date : Date
- category : String
- User : User
- initiatedBy : User
- referenceId : String
- mode : TxnMode
+ record() : void
+ categorise (category) : void
records
1
1
triggers
1
1
Employee
- empId : String
- designation : Enum
- departmentId : String
- joinDate : Date
- exitDate : Date
- payrollConfig : PayrollConfig
+ getPayslips (year) : List <Payslip>
+ getLeaveBalance() : int
+ updatePersonalDetails (details) : void
1
1
has
1
1
Payslip
- payslipId : String
- emp : Employee
- month : String
- gross : Float
- deductions : Float
- netPay : Float
- status : PayslipStatus
- approvedBy : User
+ generatePayslip() : void
+ approve (approverId) : void
+ reject (reason) : void
+ triggerFinanceTransaction() : Transaction
+ downloadAsPDF() : File
received by
*
PayRollConfig
- configId : String
- emp : Employee
- basic : Float
- allowances : Map <String, Float>
- deductions : Map <String, Float>
- effectiveFrom : Date
+ computeGrossSalary() : Float
+ computeNetSalary() : Float
+ updateAllowance (type, amount) : void
+ updateDeduction (type, amount) : void

Department
- departmentId : String
- name : String
- code : String
- headFaculty : Faculty
+ getMembers () : List <User>
+ getFaculty () : List <Faculty>
+ getStudents () : List <Student>
offers
1
*
Course
- courseId : String
- courseCode : String
- title : String
- credits : int
- type : CourseType
- program : List < ProgramType >
- syllabus : String
- instructor : List < Faculty >
- semester : String
+ getEnrolledStudents () : List<Student>
+ getAttendanceRecords() : List < AttendanceRecord>
+ getExams : List < Exam>
+ getTimetableSlots () : List< TimetableSlot>
+ computeAttendancePercentage (StudentId) : Float
Broadcast
- broadcastId : String
- sender : User
- audience : AudienceType
- message : String
- sentAt : dateTime
+ sendToAll () : void
+ sendToBatch (BatchId) : void
+ sendToDepartment (deptId) : void
+ schedule (dateTime) : void
employs
targeted at
0..*
Faculty
- facultyId : String
- specialisation : String
- location : String
+ getCoursesTaught (sem) : list <Course>
+ getStudents () : List < Student>
teaches
advises
houses
targeted at
Batch
- batchId : String
- admissionYear : Int
- students : List < Student>
- allowedElectives : List < Course>
+ getStudents () : List < Student>
+ addStudent (Id) : void
+ removeStudent (Id) : void
+ transferStudent (Id, bId) : void
+ getCourses (semester) : List <Course>
has
contains
Student
- StudentId : String
- batchId : String
- programType : ProgramType
- cgpa : Float
- advisor : Faculty
- coadvisor : Faculty
- guardian : List < Guardian>
- emergency contact : List <String>
- p. Education : List < String>
+ getCourse (sem) : List < Course>
+ register (CId) : Boolean
+ getAttSummary (CId) : List <O>
+ getExamResults (sem) : List<Marks>
+ viewFeeStatus () : List < FeeRecord>
+ computeCGPA() : Float
has
Enrollment
- enrollmentId : String
- student : Student
- course : Course
- credits : int
- finalGrade : String
+ enroll () : Enrollment
+ drop () : void
+ complete() : Enrollment
+ getFinalGrade (eId, sId) : String
receives
Marks
- marksId : String
- exam : Exam
- student : Student
- marks : Float
- grade : String
+ computeGrade () : Str
+ updateMarks (M) : void

Grading System
- systemId: string
- method: GradingMethod
- gradeBoundaries: List<int>
- passMark: Float
+ assignGrades (marks, max): String
+ computeGPA (marks List): Float
+ IsPass (marks): Bool
Course
- courseId: String
- courseCode: String
- title: String
- credits: int
- type: CourseType
- program: ProgramType
- syllabus: String
- instructor: List<Faculty>
- semester: String
+ getEnrolledStudents (): List<Student>
+ getAttRecords (date): List<AttendanceRecord>
+ getExams: List<Exam>
+ getTimetableSlot: List<TimetableSlot>
+ computeAttpercentage (studentId): Float
Exam
- examId: String
- course: Course
- room: Room
- start: Time
- type: ExamType
- date: Date
- duration: Int
- maxM: Float
- gradingsystem: GradingSystem
- batch: List<Batch>
- question_answers: List<pair<String, String>>
+ scheduleExam (det): Void
+ enterMarks (sId, Marks): Marks
+ computeGrades (): List<M>
+ generateMarksheet (batchId): Map<String, Float>
+ getStatistics: List<Obj>
+ InputQAns (string, string): V
+ getAns (): Pair<String, S>
Marks
- marksId: String
- exam: Exam
- student: Student
- marks: Float
- grade: String
+ computeGrade (): String
+ updateMarks (newtMarks): void
Room
- roomId: String
- name: String
- capacity: Int
- type: RoomType
- building: String
+ checkavailability (slot): Bool
+ book (sid, details): TimetableSlot
+ getTimetable (): List<TimetableSlot>
Timetable Slot
- slotId: String
- start: Time
- course: Course
- end: Time
- day: Day
- room: Room
- semester: String
+ checkconflict (slot): Bool
+ reschedule (newd, newt): v
+ getWeeklyLoad (Iid): int
+ alert on limit breach (type): Void
1
uses
*
contains entries of
*
1
has
*
1
is scheduled in
1..*
is in
1
1
hosts
*
1


Student
- studentId: String
- batchId: String
- program: ProgramType
- cgpa: Float
- advisor: Faculty
- coadvisor: Faculty
- admissiondate: Date
- guardians: List<Guardian>
- Emergencycontact: List<String>
- P.Education: List<String>
+ getEnrolledCourse (sem): List<Courses>
+ register (c): Boolean
+ getAttSummary (cid): List<O>
+ getExamResults (sem): List<Marks>
+ viewFeestatus (): List<FeeRecords>
+ computeCGPA (): Float
AttendanceRecord
- attendanceId: String
- Student: Student
- course: Course
- date: Date
- Status: AttendanceStatus
- remarks: String
+ markPresent () : void
+ markAbsent () : void
+ markLeave (r): void
+ generatedailyReport (cid, date) : List<Attendances>
+ generateMonthly Report (cid, month) : List<Attendances>
1
*
has attendance in
Message
- messageId: String
- sender: User
- recipients: List<User>
- subject: String
- body: String
- sentAt: DateTime
- readAt: DateTime
- attachments: List<File>
+ send () : void
+ markasRead (uId): void
+ reply (body): Message
+ forward (uid): Message
*
1
has reply
User
- UserId: String
- password: String (Hash)
- role: List<Role>
- IsActive: Boolean
- Lastlogin: DateTime
- Language: String
- timezone pref: String
+ login (creds): String
+ logout: void
+ changepwd (old, new): Boolean
+ haspermission (module, action): Boolean
+ getNotifs (): List<O>
+ markAlertRead (): v
+ applyforleave (appl.): LeaveApplication
1
authored by
*
1
posts
*
NewsItem
- newsId: String
- title: String
- author: User
- publishedat: DateTime
- tag: List<Str>
- ispublished: Boolean
+ publish (): Void
+ unpublish (): void
+ editcontent (title, body): void
+ searchby Tag (tag): List<NewsItem>
+ getComments (): List<NewsComments>
has
1
*
NewsComment
- commentId: String
- news: NewsItem
- user: User
- text: String
- postedAt: DateTime
- isdlt: Boolean
+ post (): void
+ moderateand delete (): void
+ flag (reason): void

<<enumeration>>
GradingMethod
ABSOLUTE
RELATIVE

<<enumeration>>
ApprovalStatus
APPROVED
PENDING
REJECTED

<<enumeration>>
PaymentStatus
PAID
PENDING
CANCELLED

<<enumeration>>
CourseType
H1
H2
H
FULL

<<enumeration>>
TxnMode
UPI
RTGS
NET-BANKING
BANK-LOAN
DEMAND-DRAFT
CHEQUE
CARD

<<enumeration>>
AccessLevel
VIEW
EDIT
DELETE

<<enumeration>>
AudienceType
BATCH
DEPARTMENT
ALL
HOSTEL1
HOSTEL2
FACULTY

<<enumeration>>
ProgramType
CSE
ECE
CSD
ECD
CLD
CHD
CGD
CND
PG
MTech

<<enumeration>>
InterviewMode
ONLINE
OFFLINE

<<enumeration>>
InterviewOutcome
RECRUITED
NOT RECRUITED

<<enumeration>>
Module
DASHBOARD
ADMISSION
STUDENT DETAILS
EXAMINATIONS
MANAGE USERS
HR
ATTENDANCE
FINANCE
MESSAGES
TIME TABLE
MANAGE NEWS

<<enumeration>>
AttendanceStatus
PRESENT
ABSENT
APPROVED_LEAVE
UNMARKED

<<enumeration>>
RoomType
Science Lab
Computer Lab
100-C Rooms
300-C Rooms

<<enumeration>>
ExamType
QUIZ1
MIDSEM
QUIZ2
ENDSEM

<<enumeration>>
FeeCategory
HOSTEL
MESS
TUITION

<<enumeration>>
Designation
PROFESSOR
ASSOCIATE-PROFESSOR
DEAN-OF-ACADEMICS
LAB-TECHNICIANS
REGISTRAR
MAINTENANCE-WORKERS
SECURITY-PERSONS

<<enumeration>>
PayslipStatus
APPROVED
PENDING
SCHEDULED

<<enumeration>>
LeaveType
MEDICAL
ACADEMIC
EMERGENCY
VACATION

<<enumeration>>
TxnType
TUITION FEE
FINANCIAL AIDS
REFUNDS

<<enumeration>>
DayEnum
MONDAY
TUESDAY
WEDNESDAY
THURSDAY
FRIDAY
SATURDAY
SUNDAY